GATE

ALL BRANCHES



General Aptitude

ANALYTICAL APTITUDE

Lecture No.- 05



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Recap of Previous Lecture



Topic Problem Solving

Topics to be Covered



Topic-1 Cubes

Topic-2 Dices

[MCQ]

#Q. Five boys took part in a race. Ronit finished before Mohit but behind Gopal. Abbas finished before Sailesh but behind Mohit. Who won the race?

Assignment

A Ronit

B Gopal

C Mohit

D Abbas

Abbas
Sailesh

Gopal
Ronit
Mohit
Abbas
Sailesh

[MCQ]

#Q. In a photograph Shyam is to the left of Madan. Mary is to the right of George, Karim is in between Shyam and Mary. Who is at the corner?

Assignment

George Mary Karim Shyam Madan

A Shyam

B Mary

C George

D Karim

George Mary

[MCQ]

#Q. Raju and Manoj are ranked 14 and 15 respectively from the top in a class of 30 students. What will be their respective ranks from the bottom.

Assignment

Manoj $\Rightarrow 30-15+1 = 16^{th}$

Raju $\Rightarrow 30-14+1 = 17^{th}$

Raju (14th from top) — 17th from bottom

Manoj (15th from top) — 16th from bottom

A. 15th and 16th

B. 16th and 15th

C. 17th and 16th

D. 18th and 17th

[MCQ]

#Q. There are six teachers A, B, C, D, E and F in a school. Each of the teachers teaches two subjects, one compulsory subject and other optional subject. D's optional subject was History while three others have it as compulsory subject. E and F have Physics as one of their subjects. F's compulsory subject is Mathematics which is an optional subject of both C and E. History and English are A's subjects but in terms of compulsory and optional subjects, they are just reverse of those of D's. Chemistry is an optional subject of only one of them. The only female teacher in the school has English as her compulsory subject.





[MCQ]

#Q. What is C's compulsory subject?

History

3 ⇒ Comp Hist

E | E

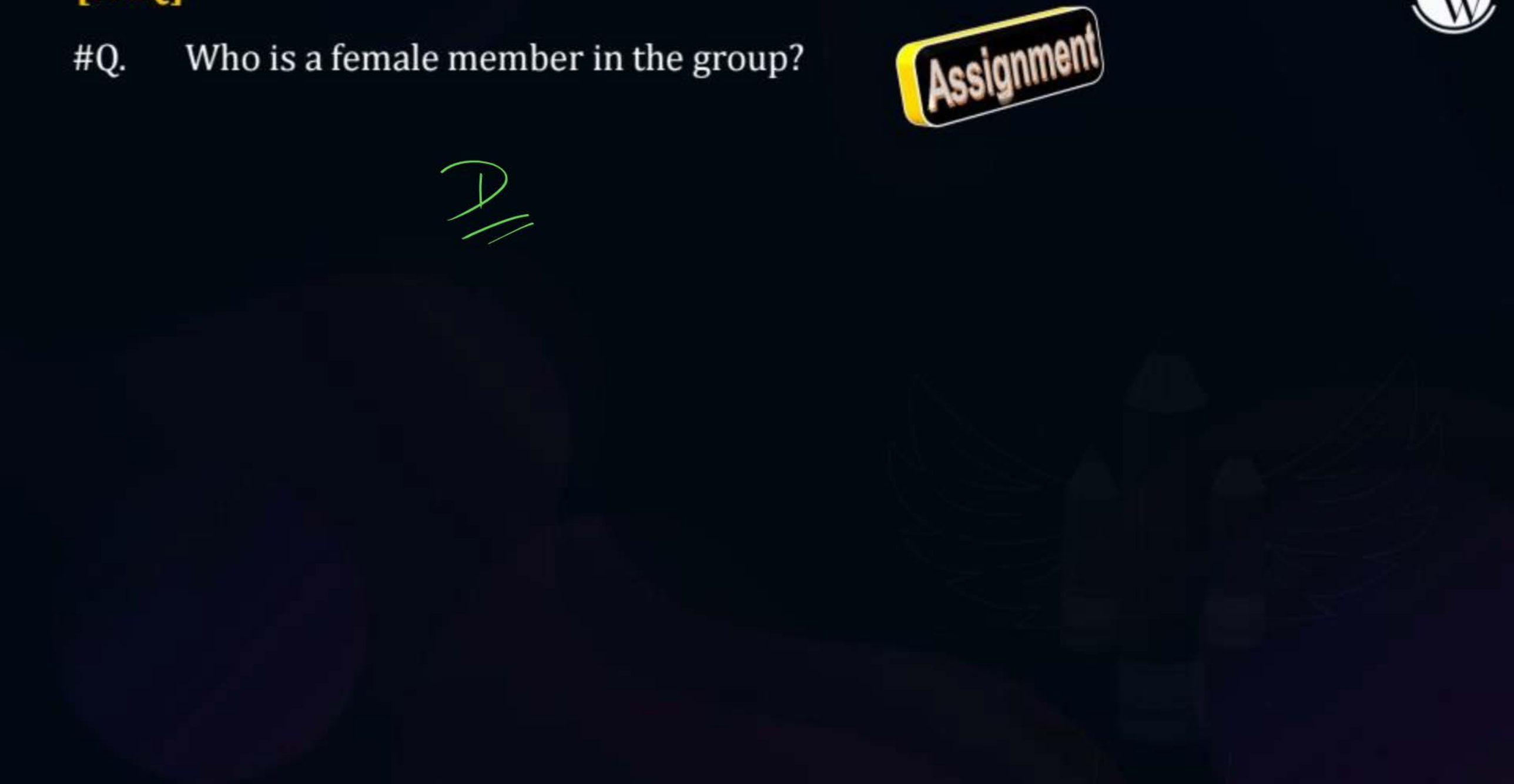
Physics

Assignment

	Comp	Optional
A	Hist	English
B	Hist	Chemistry
C	Hist	Maths
D	English	Hist
E	Physics	Maths
F	Maths	Physics

[MCQ]

#Q. Who is a female member in the group?



[MCQ]

#Q. Disregarding which is the compulsory which is the optional subject, who has same two subject combination as F?

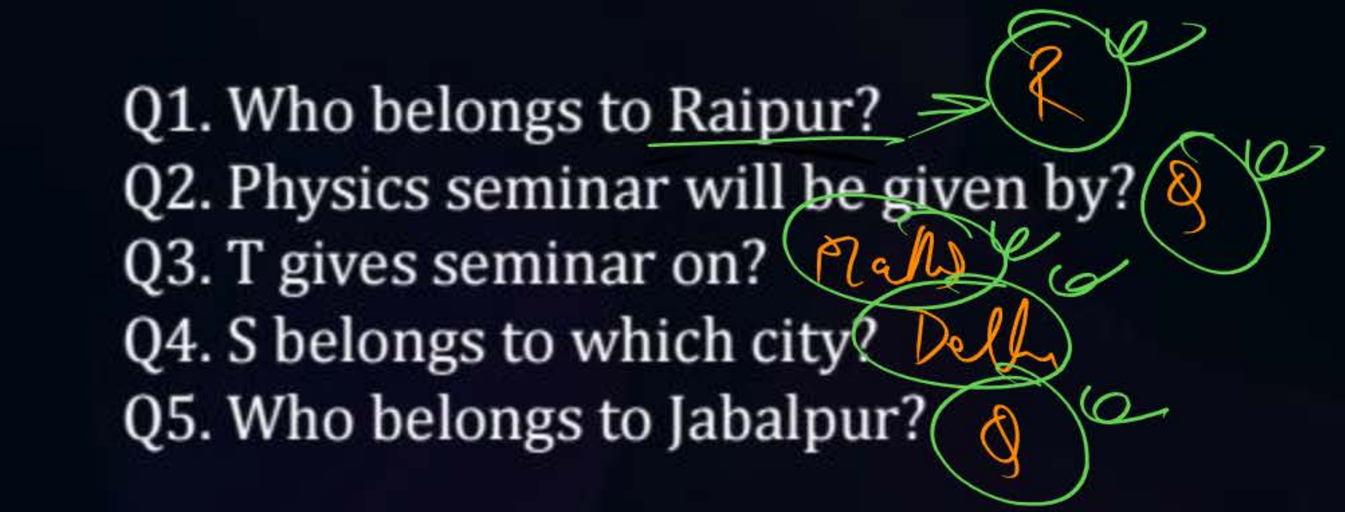


E

[MCQ]

#Q. Five persons P, Q, R, S and T belongs to five different cities Jabalpur, Bhopal, Hyderabad, Raipur and Delhi. They are invited to deliver a seminar on different subject Viz., Chemistry, Physics, Mathematics, History and Economics but not necessarily in the same order.

P who belongs to Bhopal gives seminar on Economics. The person who gives seminar on Physics belongs to Jabalpur. T does not give seminar on Chemistry or History but belongs to Hyderabad. S does not belongs to Raipur but gives seminar on Chemistry and History is the option of R.



Q1. Who belongs to Raipur?
Q2. Physics seminar will be given by?
Q3. T gives seminar on?
Q4. S belongs to which city?
Q5. Who belongs to Jabalpur?

- Five persons P, Q, R, S and T belongs to five different cities Jabalpur, Bhopal, Hyderabad, Raipur and Delhi. They are invited to deliver a seminar on different subject Viz., Chemistry, Physics, Mathematics, History and Economics but not necessarily in the same order.
- P who belongs to Bhopal gives seminar on Economics.
- The person who gives seminar on Physics belongs to Jabalpur.
- T does not give seminar on Chemistry or History but belongs to Hyderabad.
- S does not belongs to Raipur but gives seminar on Chemistry and History is the option of R.

	CITY	SUBJECT
P	Bhopal	Econ
Q	Jabalpur	Physics
R	Raipur	Hist
S	Delhi	Chem
T	Hyderabad	Maths

CITY	JABALPUR	BHOPAL	HYDERABAD	RAIPUR	DELHI
SUBJECT	Physics				

[MCQ]

#Q. On a shelf are placed six volumes side by side labeled A, B, C, D, E and F. Three volumes B, C and E have green covers while the other volumes have yellow covers. A, D and B are new volumes while the rest are old volumes. A, C and B are law reports, while the rest are Gazetteers.

Assignment



[MCQ]

#Q. Which volume is new, yellow covered and a Gazetteer?

A E

B D

C C

D A

[MCQ]

#Q. Which two volumes are old Gazetteers?

A E and F

B D and B

C C

D F

[MCQ]

#Q. Which is green covered, old and a law report volume?

A E

B D

C F

D C

[MCQ]

#Q. Prashant Arora has three children – Sangeeta, Vimal and Ashish. Ashish married Monika, the eldest daughter of Mr. and Mrs. Roy. The Roys married their youngest daughter to the eldest son of Mr. and Mrs. Sharma, and they had two children name Amit and Shashi. The Roys have two more children, Roshan and Vandana, both elder to Veena. Sameer and Ajay are sons of Ashish and Monika. Rashmi is the daughter of Amit.

Assignment

Q1. What is the surname of Rashmi?

Q2. How is Sameer related to the father of Monika?

Q3. What is the surname of Sameer?

Q4. How is Mrs. Roy related to Ashish?

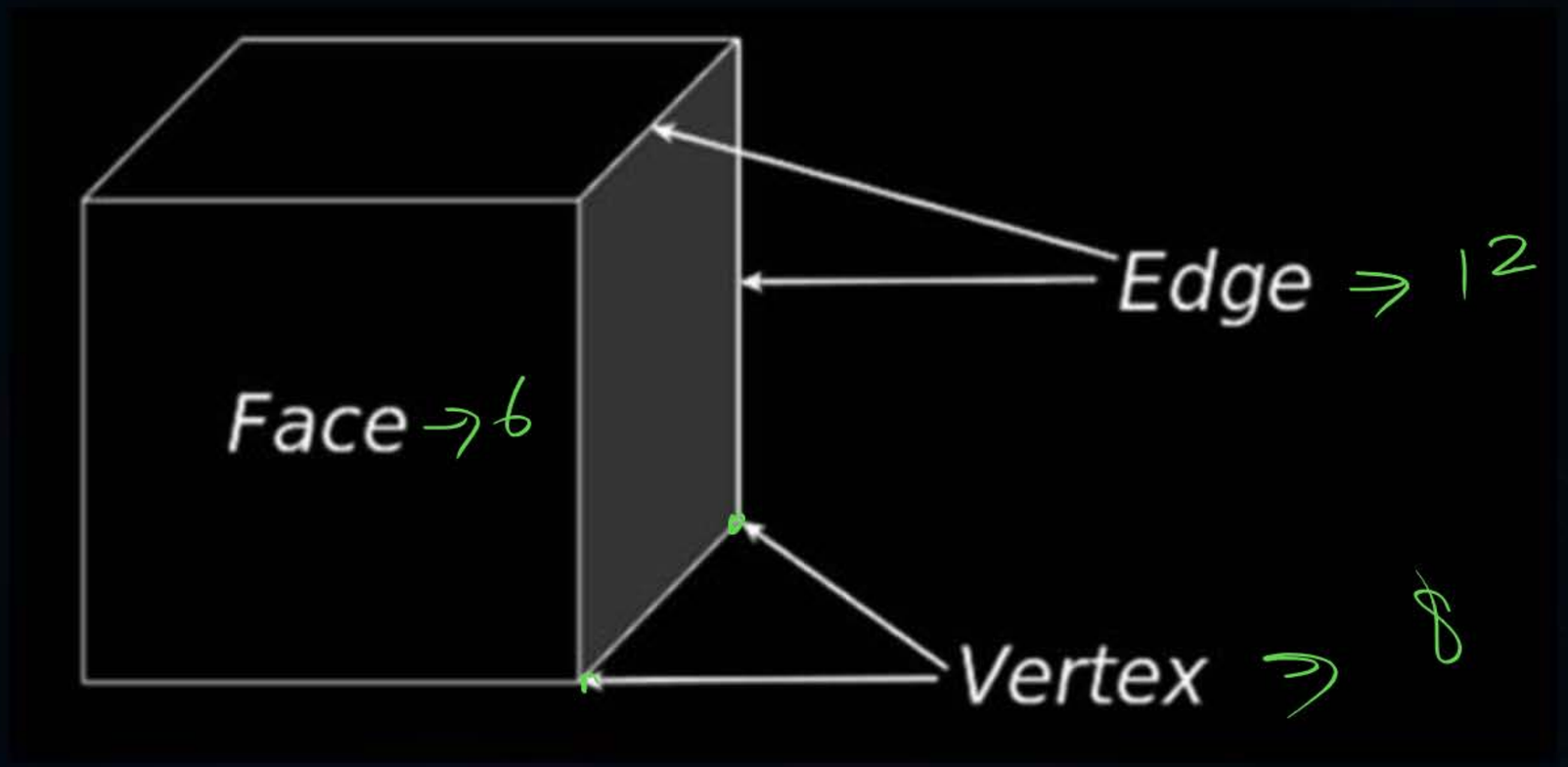
Cubes

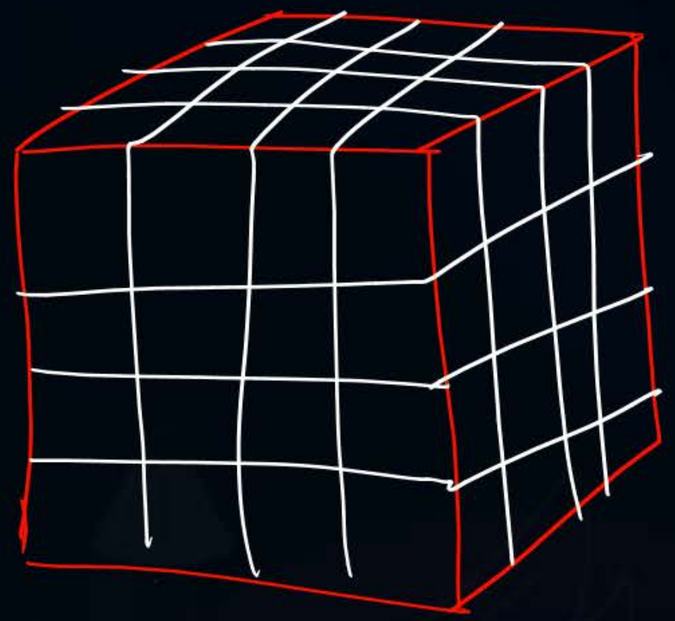


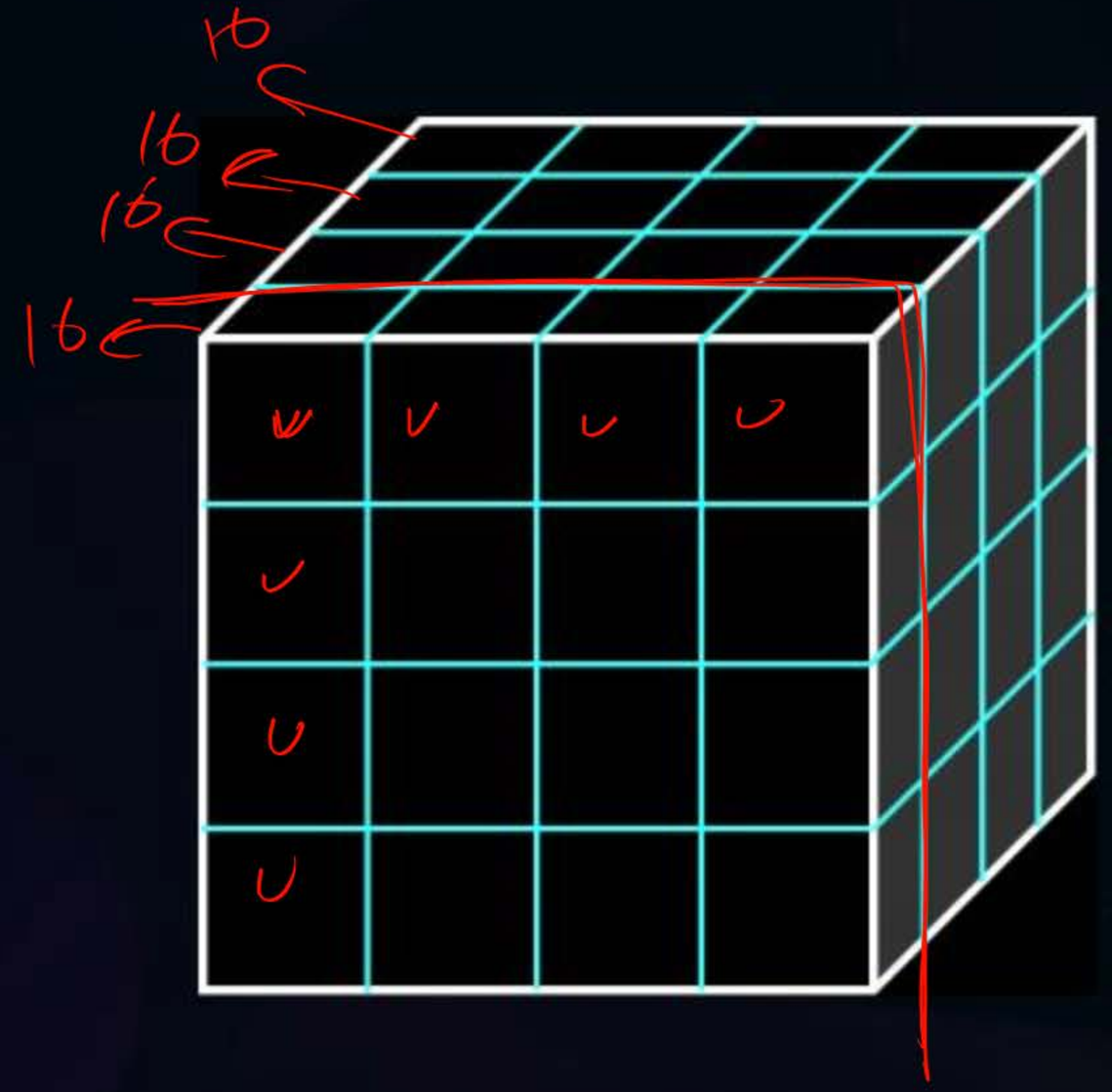
faces ⇒ 6

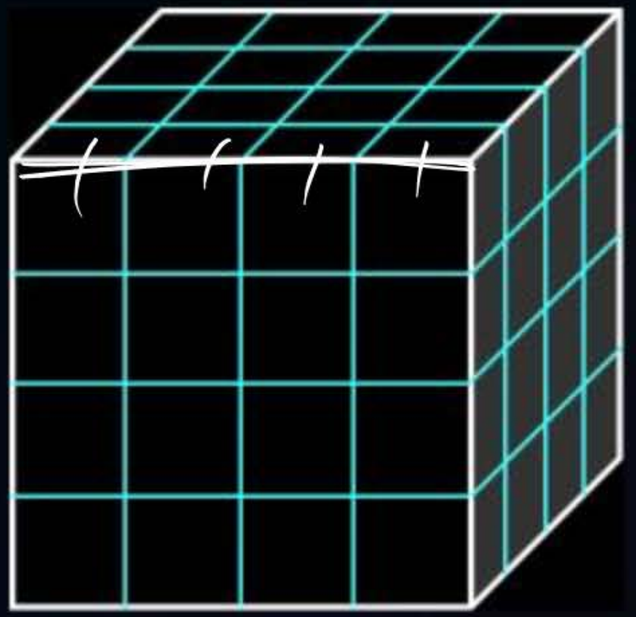
Edges ⇒ 12

Vertex ⇒ 8


Face → 6
Edge → 12
Vertex → 8




16
16
16
16
16 x 4 = 64





3

n cut

No. of cubes

$=(x+1)^3$

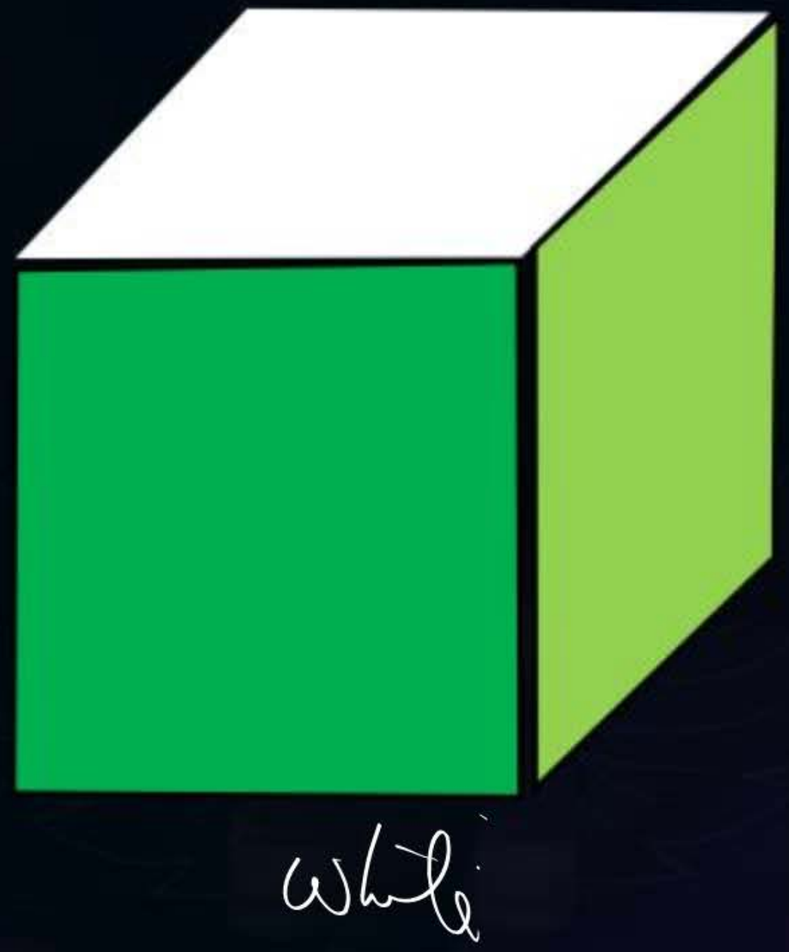
$(3+1)^3=4^3=64$

[MCQ]

#Q. A Cube was painted on one of the opposite pair faces as WHITE & remaining four faces as GREEN. Five cuts were given on each edges to form smaller Cubes of equal size.

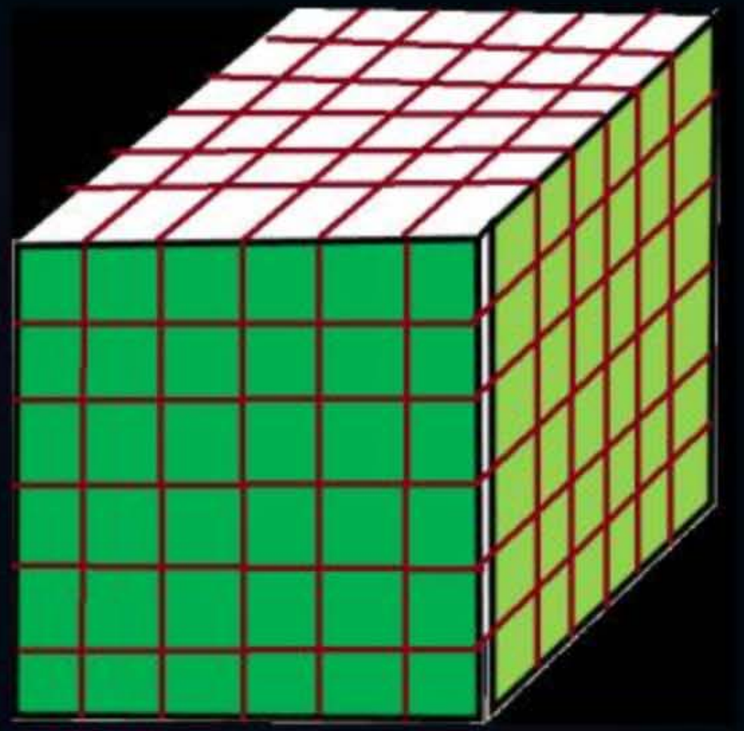
1. How many cubes will be formed?
2. How many cubes will have no face painted at all?
3. How many cubes will have only one face painted?
4. How many cubes will have only two face painted, but with same colour?
5. How many cubes will have only two face painted, but different colour?
6. How many cubes will have atleast two face painted?
7. How many cubes will have three face painted?

✓A Cube was painted on one of the opposite pair faces as white & remaining four faces as green. Five cuts were given on each edges to form smaller Cubes of equal size.



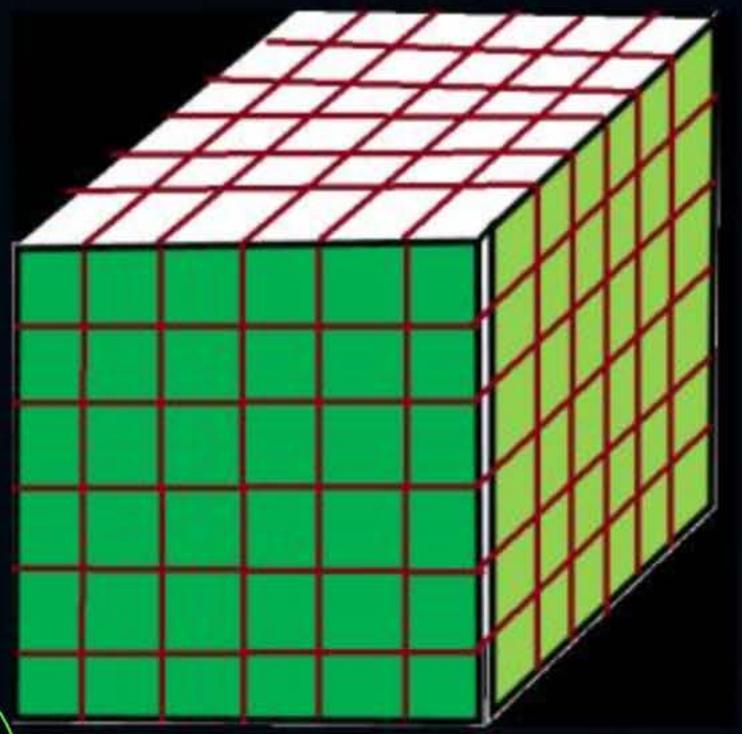
A Cube was painted on one of the opposite pair faces as white & remaining four faces as green. Five cuts were given on each edges to form smaller Cubes of equal size.





[MCQ]

#Q. How many cubes will be formed?



(5) 'x' cut

No. of cubes = $(5+1)^3$

$= 6^3 = 216$

[MCQ]

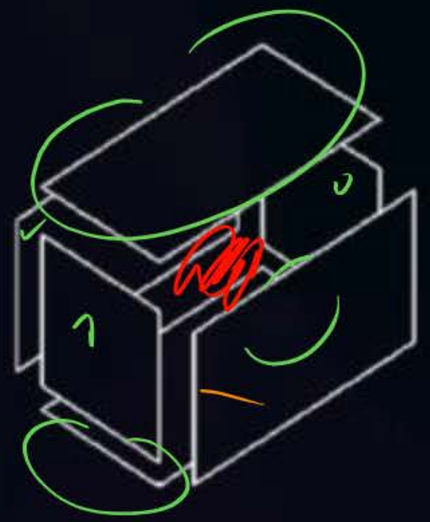
#Q. How many cubes will have no face painted at all?

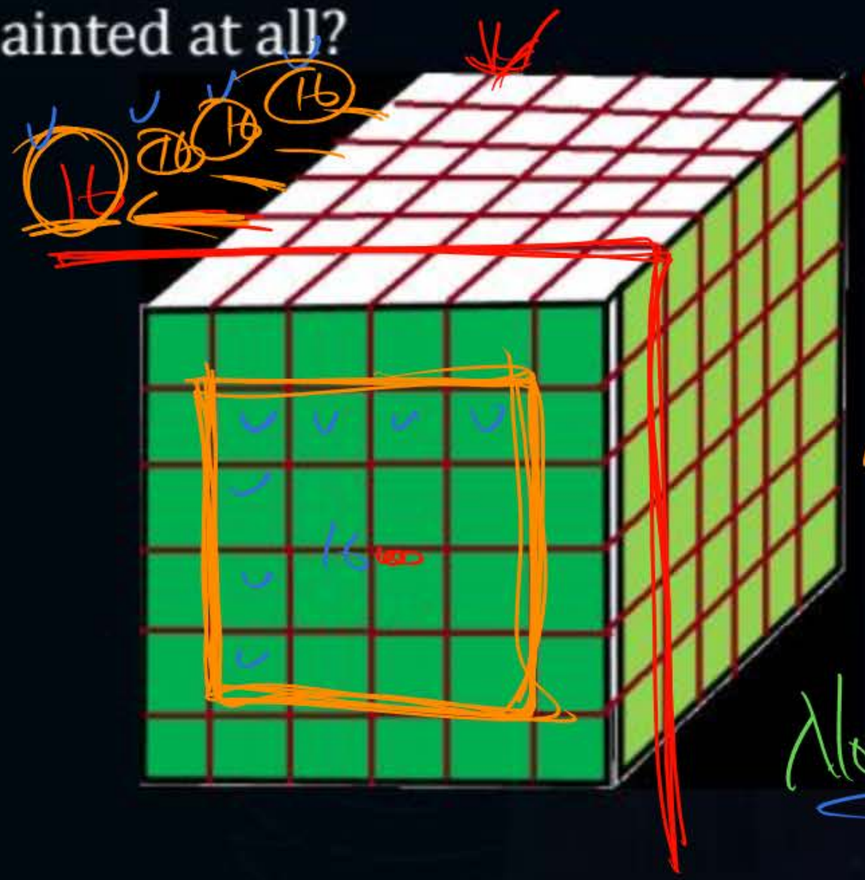
$(5-1)^3$

$=4^3$

$=64$

$16 \times 4 = 64$





5 cut

No. of cubes

$= (x+1)^3$

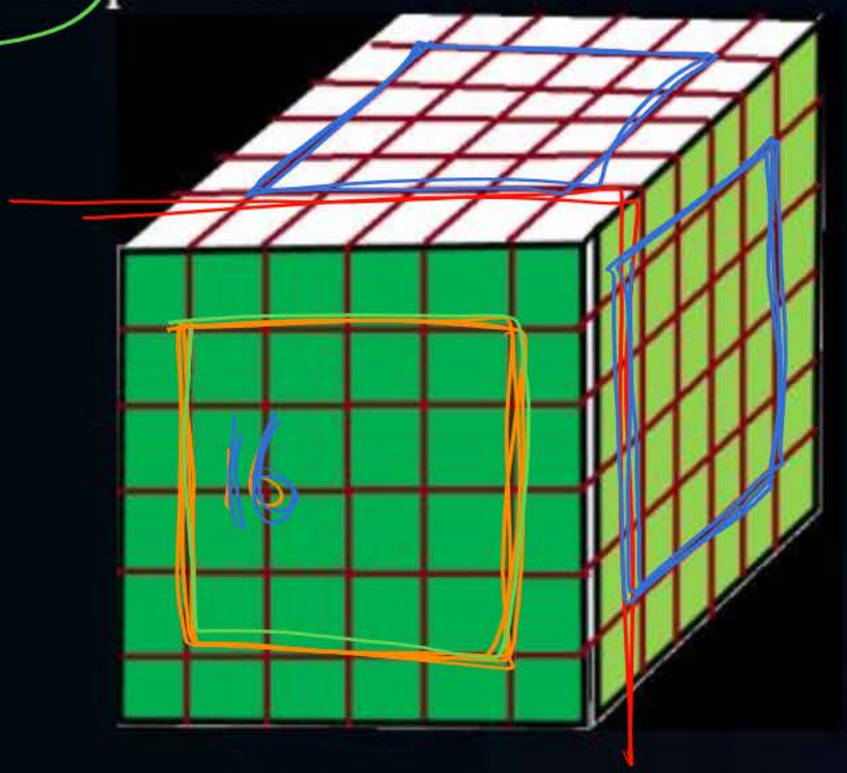
$(5+1)^3 = 6^3 = 216$

No. of cubes with no face painted

$= (x-1)^3$

[MCQ]

#Q. How many cubes will have only one face painted?



16 x 6

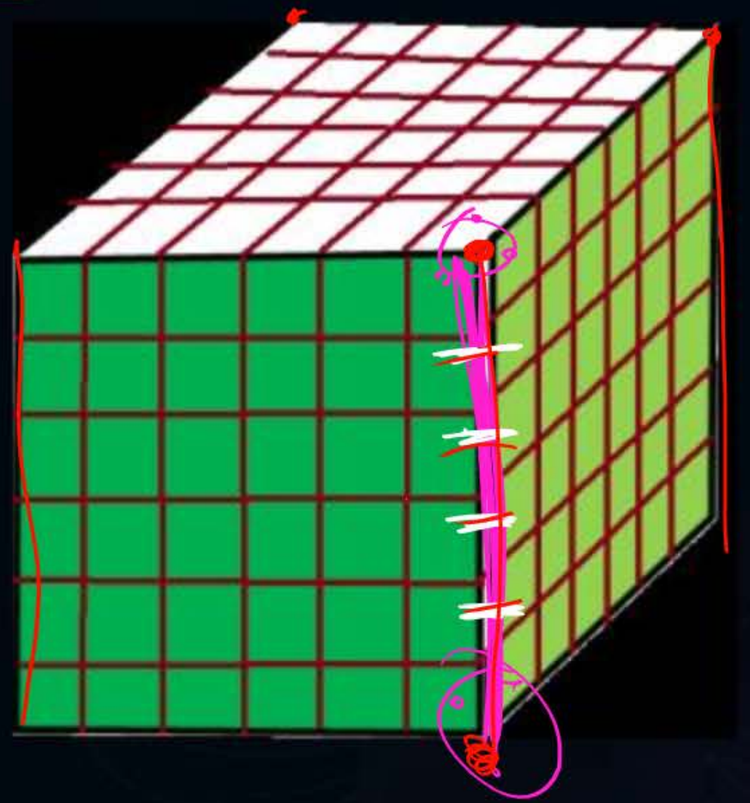
= 96

[MCQ]

12 × 4 = 48 = 32 + 16 = 48

#Q. How many cubes will have only two face painted, but with same colour?

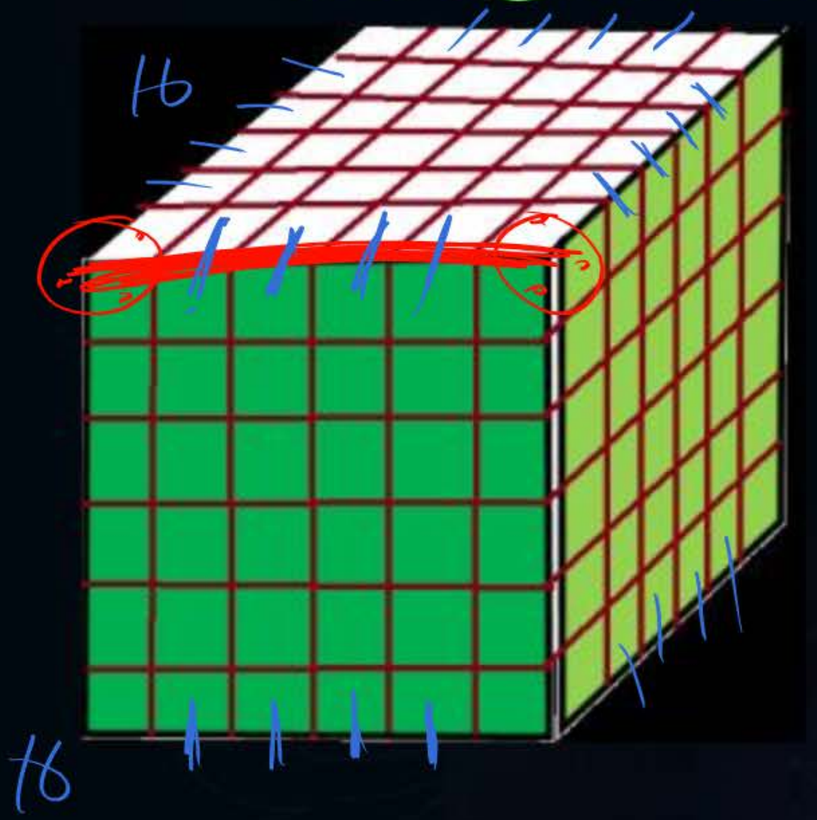
4 × 4 = 16



[MCQ]

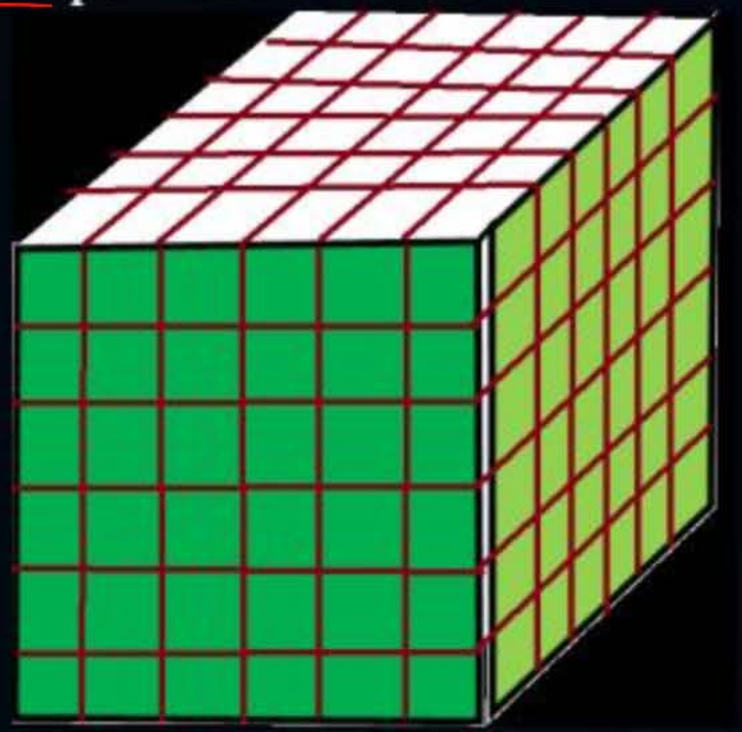
#Q. How many cubes will have only two face painted, but with different colour?

$16 + 16 = 32$



[MCQ]

#Q. How many cubes will have atleast two face painted?



Only two face

3 face paint

48 + 8

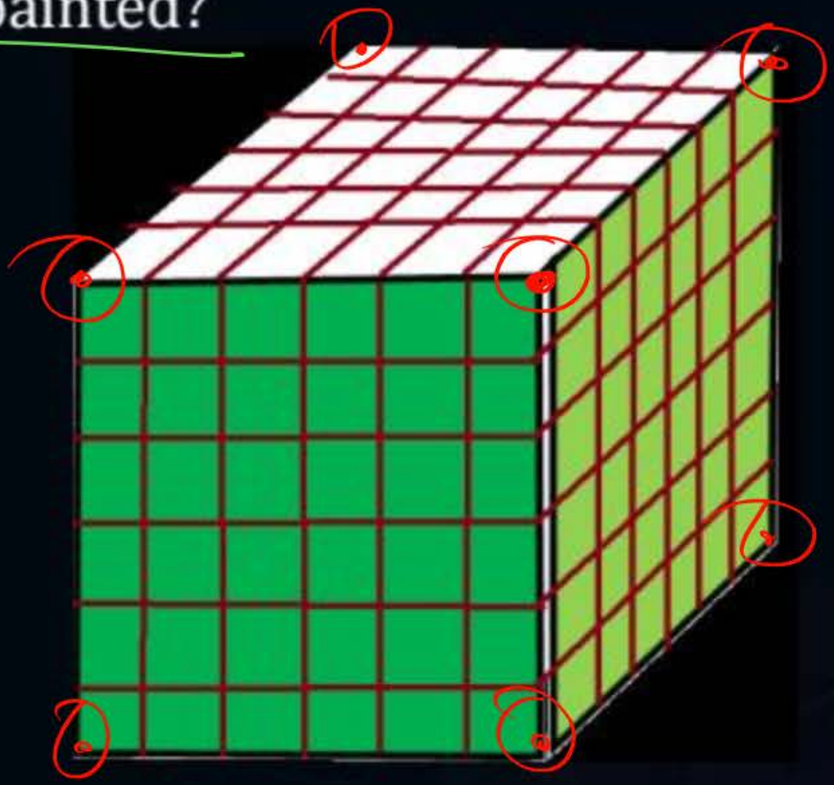
↓

56

[MCQ]

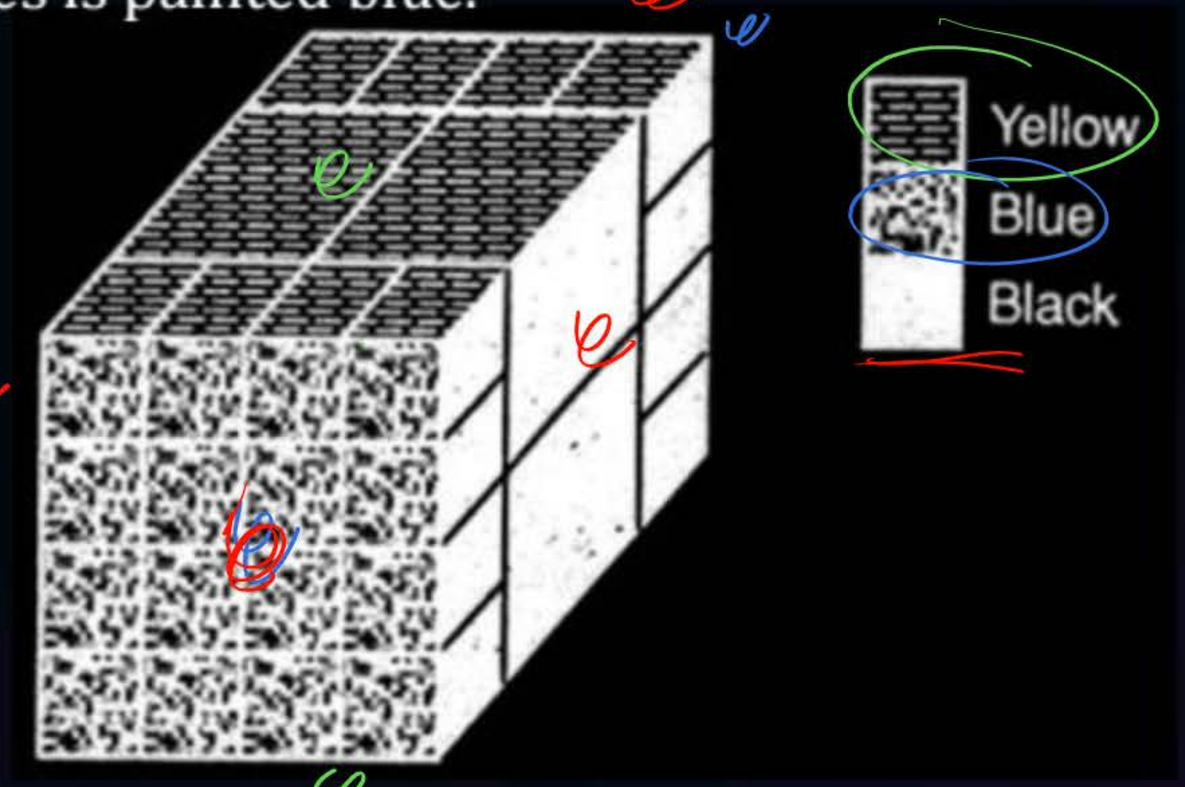
#Q. How many cubes will have three face painted?

8



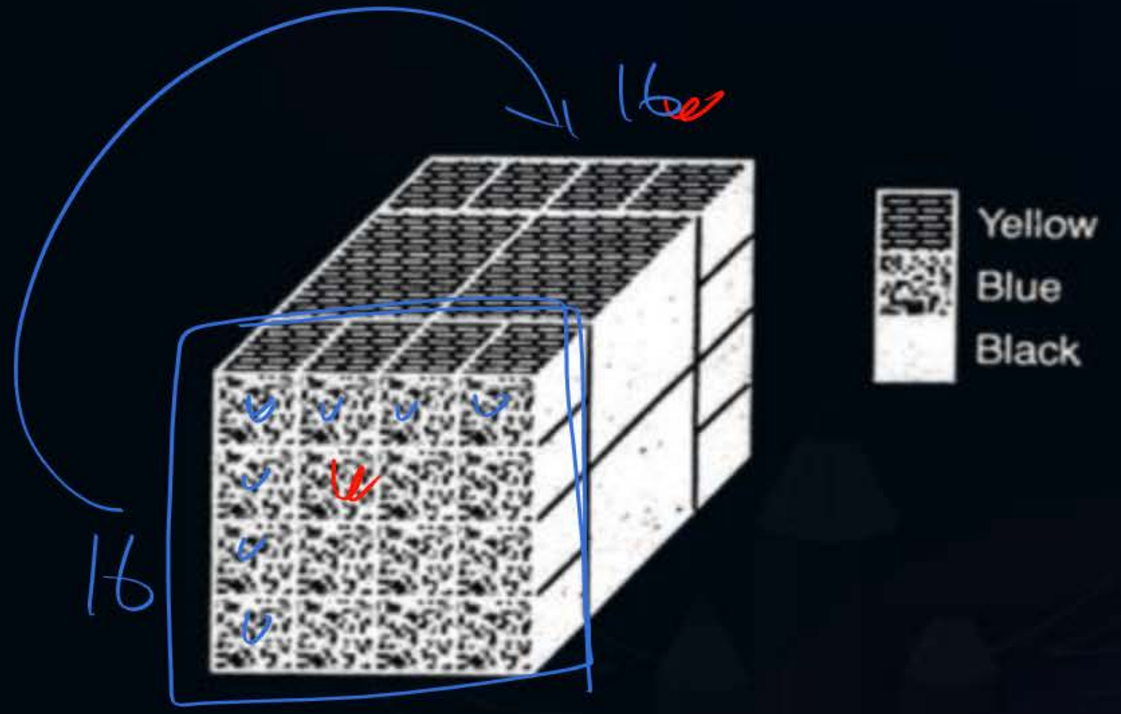
[MCQ]

#Q. A solid cube as shown in below image, has been painted yellow, blue and black on pairs of opposite face. The cube is the cut into 36 smaller cubes such that 32 cubes are of the same size while 4 others are of bigger size. Also no face of any of the bigger cubes is painted blue.



[MCQ]

#Q. How many cubes have at least one face painted blue?



16+16

=32

A 0

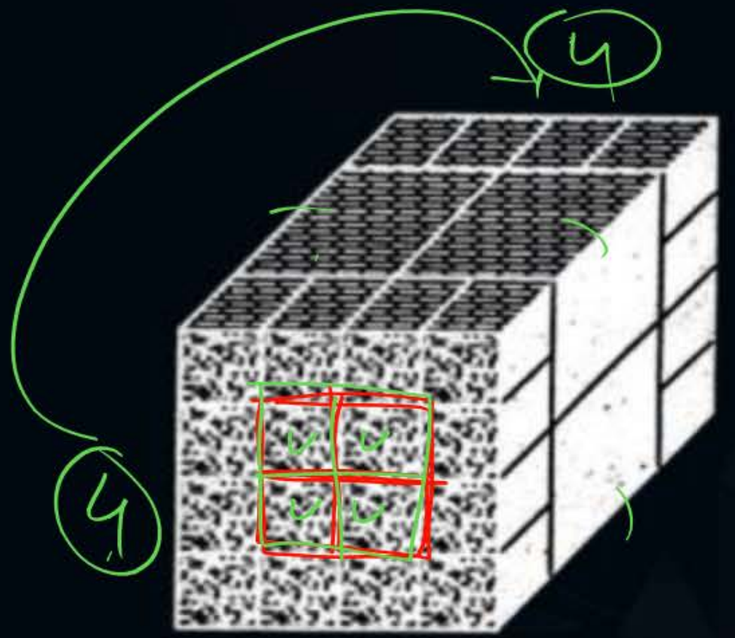
B 8

C 16

D 32

[MCQ]

#Q. How many cubes have only one face painted?



A 12

B 8

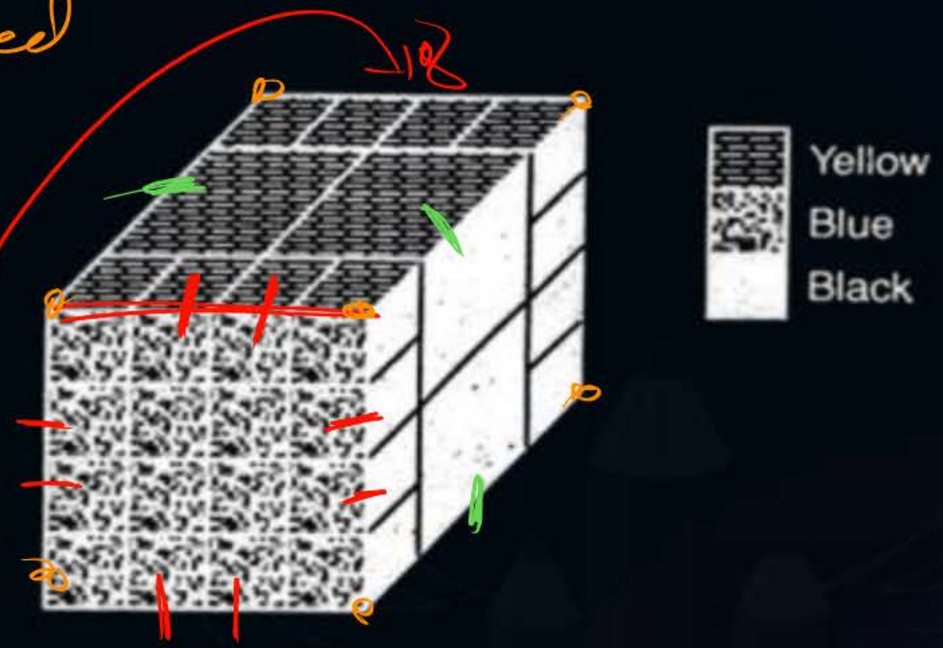
C 4

D 0

4+4=8

[MCQ]

#Q. How many cubes have two or more faces painted?



Only Two faces painted

Big ⇒ 4

Small ⇒ 16

20

Three faces ⇒ 8

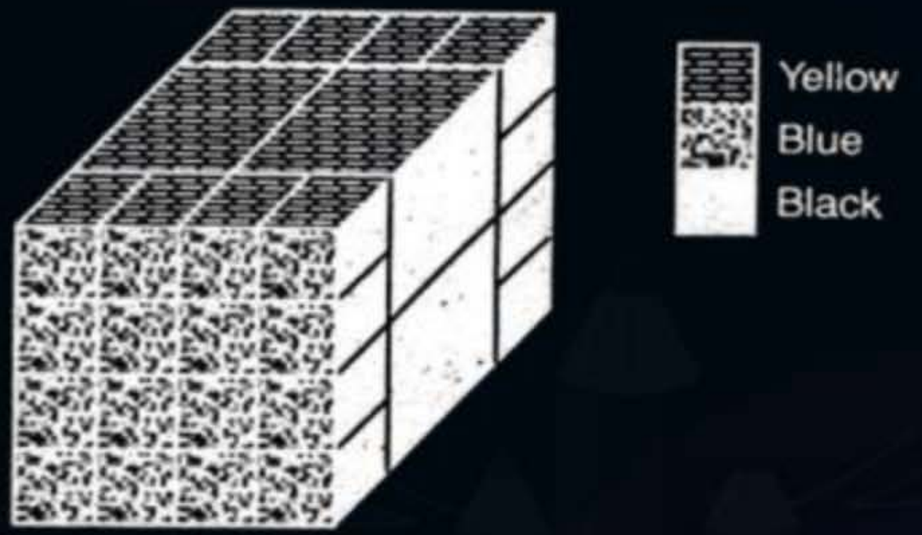
28

A	34
B	36
C	28
D	24

[MCQ]

#Q. How many cubes only three faces painted?

Assignment



A 0

B 2

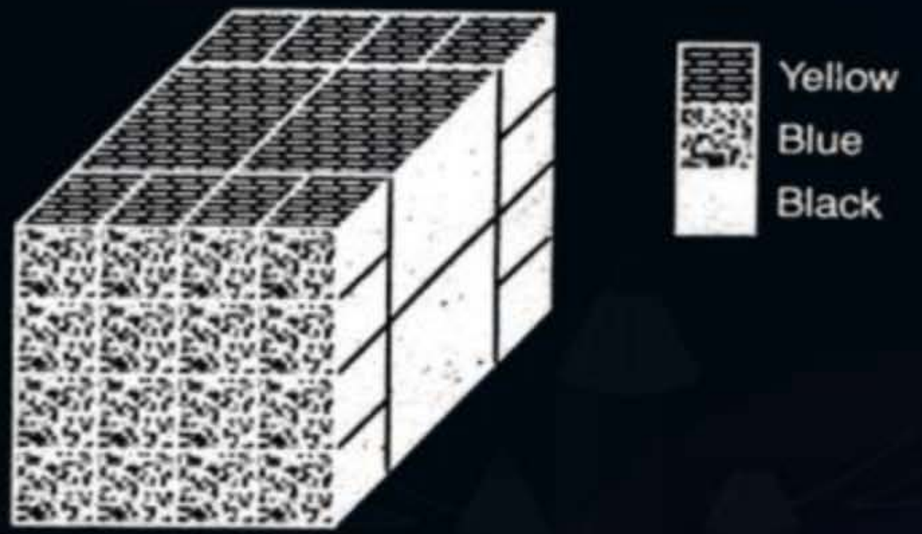
C 4

D 8

[MCQ]

#Q. How many cubes do not have any of their faces painted yellow?

Assignment



A 0

B 8

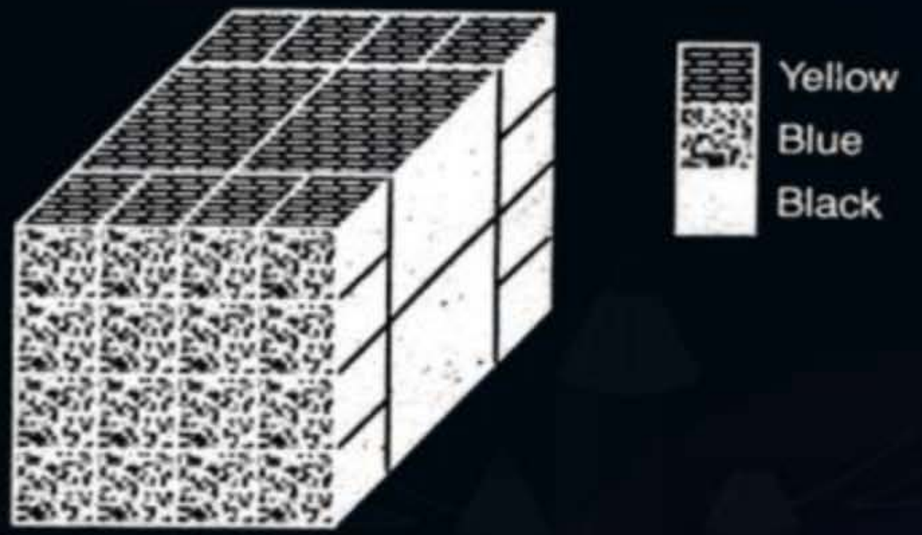
C 16

D 20

[MCQ]

#Q. How many cubes have at least one of their faces painted black ?

Assignment



A 32

B 16

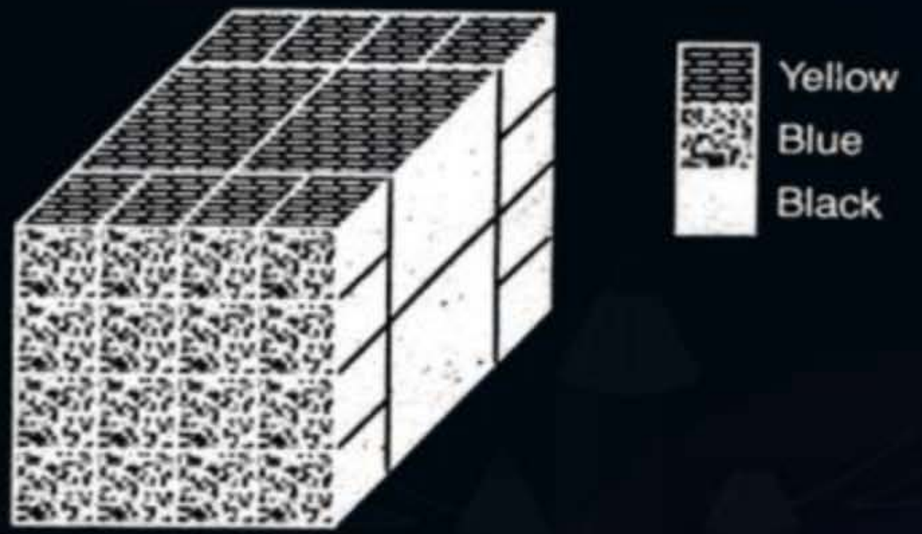
C 8

D 20

[MCQ]

#Q. How many cubes have at least one of their faces painted yellow or blue?

Assignment



A 0

B 16

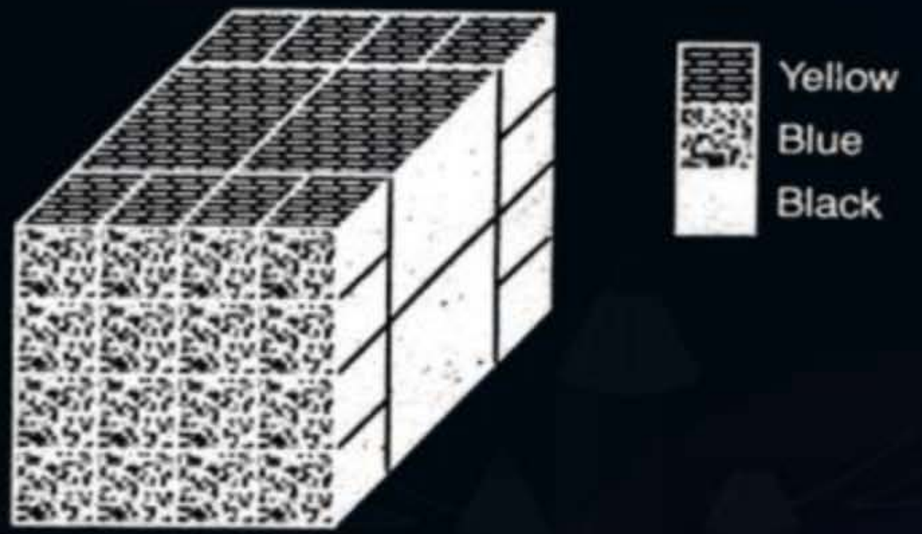
C 32

D 36

[MCQ]

#Q. How many cubes have no face painted?

Assignment



A 0

B 8

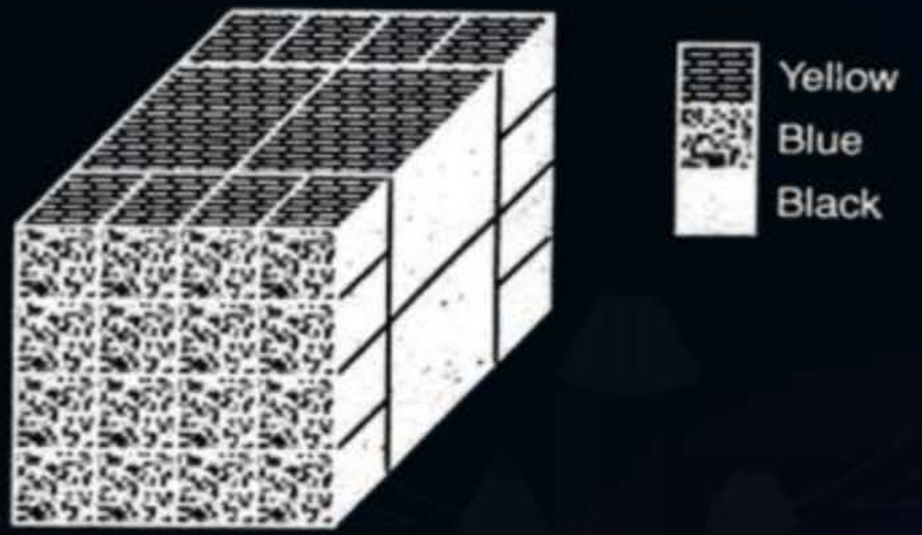
C 12

D 16

[MCQ]

#Q. How many cubes have two faces painted yellow and black respectively?

Assignment



A 16

B 12

C 8

D 0

Dice



Ludo

* Standard dice

Sum → 7

1 → 6

4 → 3

5 → 2

Standard & General Dice

- ✡ Rule ⇒ 1
- ✡ Rule ⇒ 2
- ✡ Rule ⇒ 3



(A)

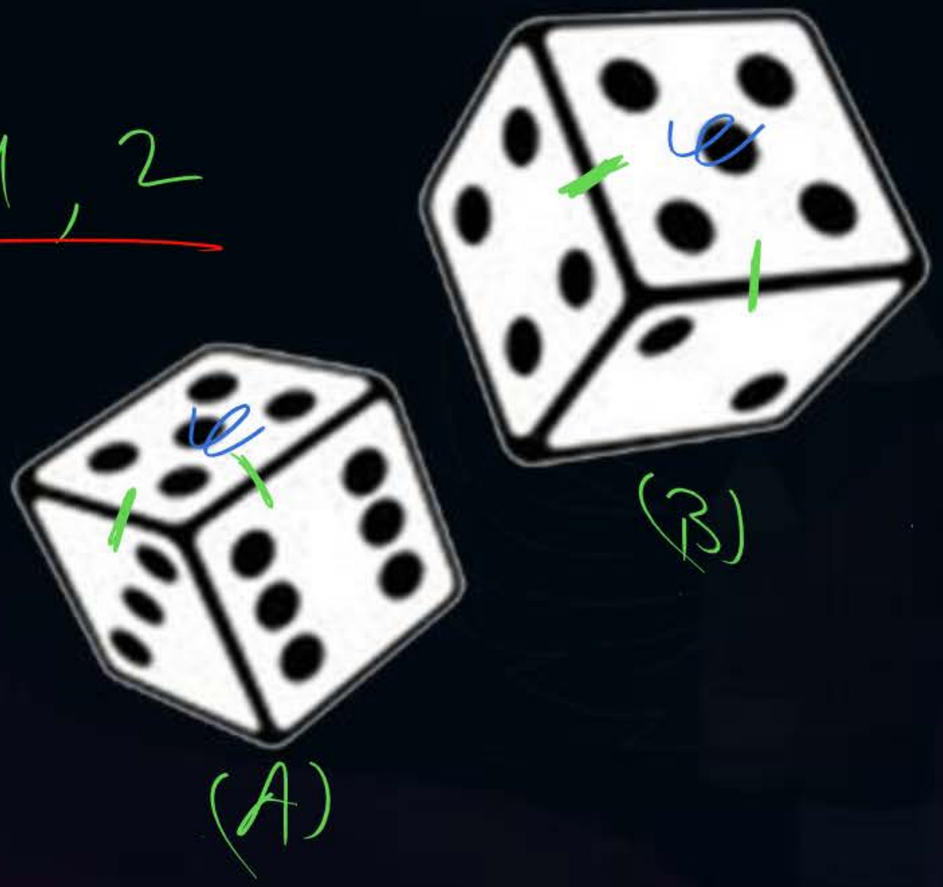


(B)

Rule 1:

5 → 6, 3, 4, 2

5 ↔ 1



Rule 2:

1 ↔ 6



Rule 3:

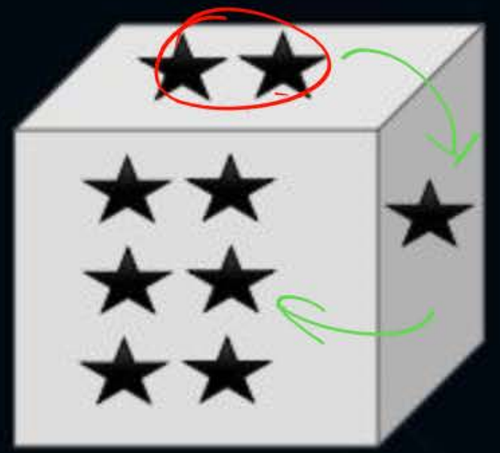
1 ↔ 3

2 ↔ ?

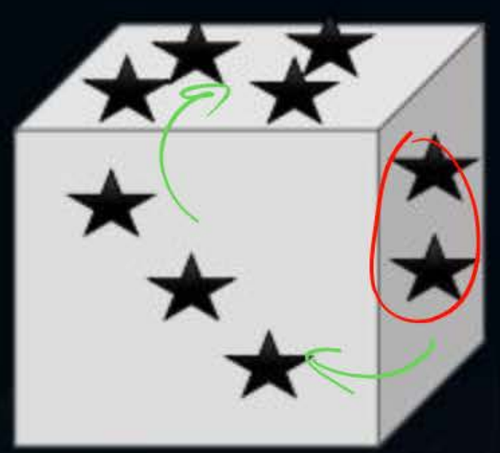
(A) 2 — 1 — 6

(B) 2 — 3 — 4

6 ↔ 4

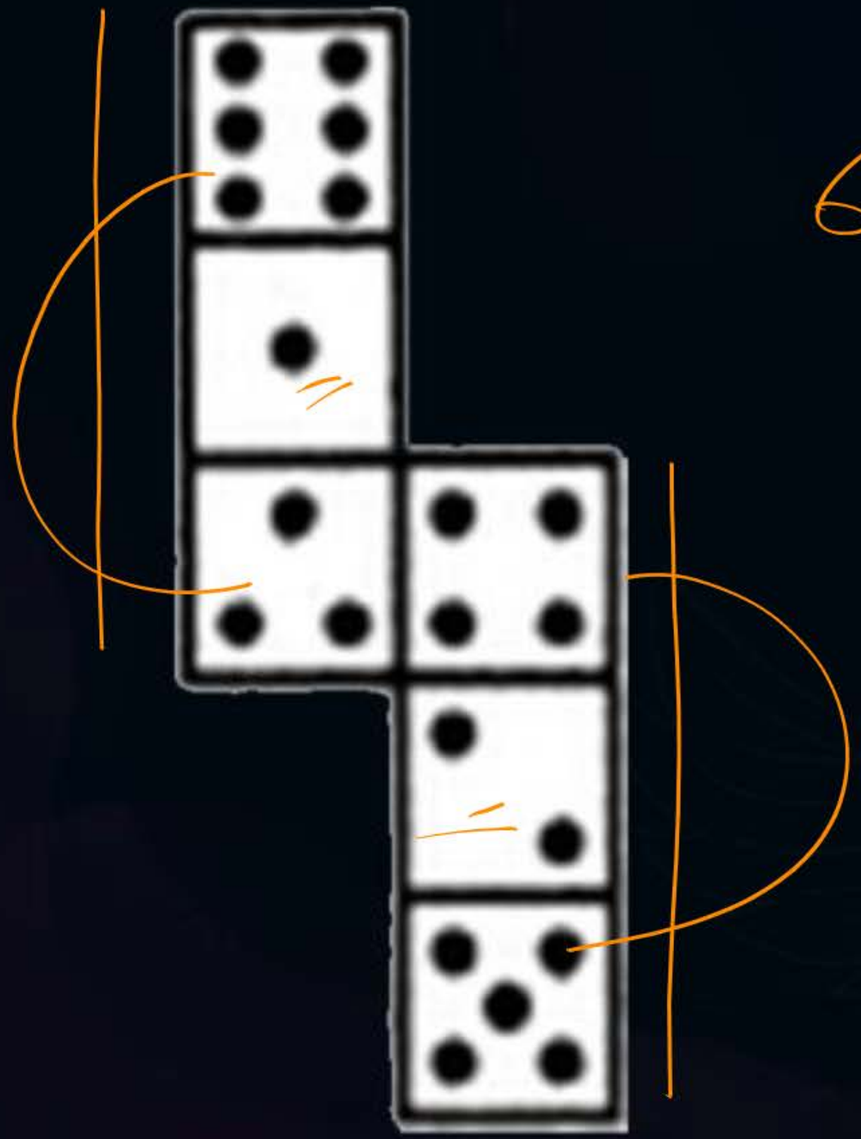


(A)



(R)

Open Dice:



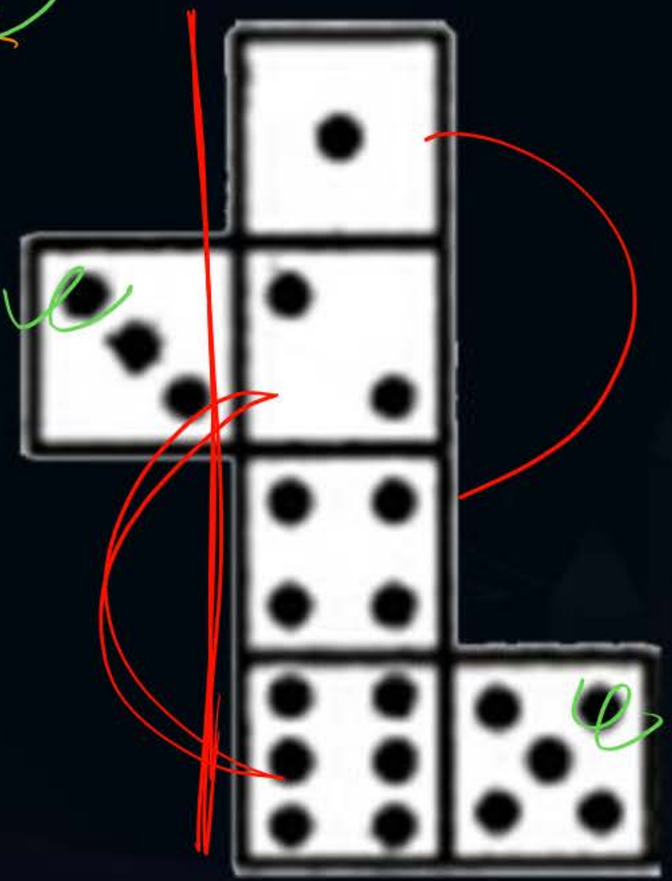
6 ↔ 3

4 ↔ 5

2 ↔ 1

[MCQ]

#Q. When the following figure is folded to form a cube, how many dots lie opposite the face bearing five dots?



A 1

B 2

C 3

D 4

[MCQ]

Assignment

#Q. Which cube can be formed by folding the given shape on the right?

A 

B 

C 

D 



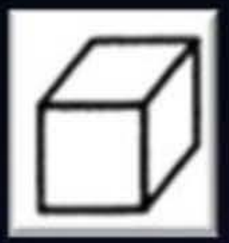
[MCQ]

Assignment

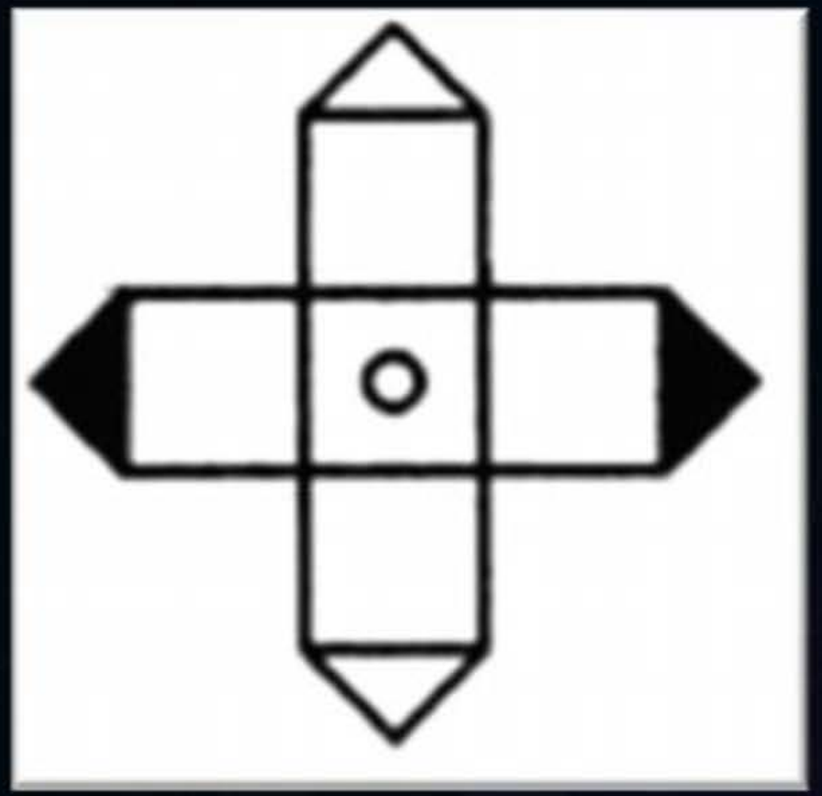
#Q. Which cube can be formed by folding the given shape on the right?

A 

B 

C 

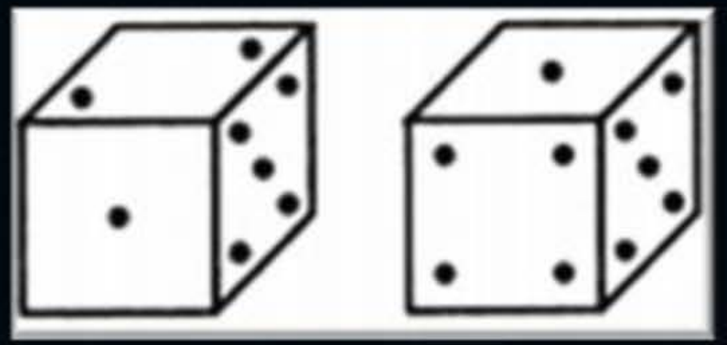
D 



[MCQ]

Assignment

#Q. Observe the dots on a dice (one to six dots) in the following figures. How many dots are contained on the face opposite to that containing four dots?



A 2

B 3

C 6

D Can't be determined

[MCQ]

Assignment

#Q. Three different positions of a dice are shown below. How many dots lie opposite 2 dots?



A 1

B 3

C 6

D 5

[MCQ]

Assignment

#Q. The six faces of a dice have been marked with alphabets A, B, C, D, E and F respectively. This dice is rolled down three times. The three positions are shown as: Find the alphabet opposite A.



A C

B D

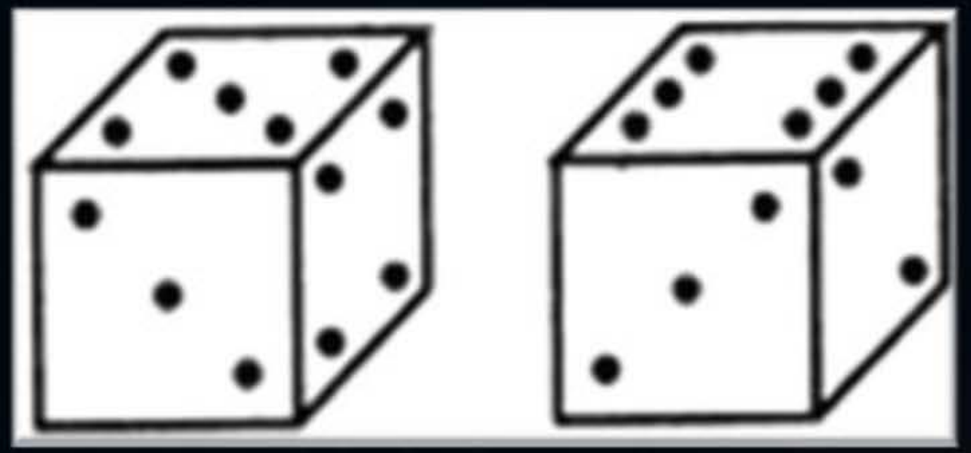
C E

D F

[MCQ]

Assignment

#Q. Two positions of a dice are shown below. If the face with 1 dot is at the bottom, then the number of dots on the top is



A 2

B 3

C 4

D 5

[MCQ]

Assignment

#Q. A cube has six different symbols drawn over its six faces. The symbols are dot, circle, triangle, square, cross and arrow. Three different positions of the cube are shown in figures X, Y, and Z.

Which symbol is opposite the dot?



(x) (y) (z)

A Circle

B Arrow

C Cross

D Triangle



2 mins Summary

Topic Cubes & Dice

THANK - YOU